## Glucose Tolerance Test for Gestational Diabetes Preparation

The patient must be educated on how to prepare for glucose tolerance test (GTT) for at least 3 days before the test. To prepare for this test, instruct the patient to consume 150 g of carbohydrates through balanced meals for a 3-day period of time. The day of the test, they should fast for at least 8 hours but no more than 16 hours. They should also avoid excessive exercise and any drinking of alcohol or smoking for the same amount of time. Inform them that they can drink water during the test, but no other food should be eaten while the test is being conducted.

## Glucose Tolerance Test for Gestational Diabetes Steps

1. Review the requisition to determine any specific requirements for this collection.
2. Introduce yourself.
3. Properly identify the patient
4. Describe the procedure's steps and approximately how long it will take to perform.
5. Perform a venipuncture to test the patient's glucose levels. When ordered, ask the patient to provide a urine sample.
6. Wait for the test results.
7. When results are in the accepted range, the testing may continue.
8. Measure out the required glucose beverage amount based on weight, which is typically 75 g for adults.
9. Ask the patient to drink the entire beverage within 5 minutes.
10. Record the time the patient finishes the drink. This is the beginning time for the test.
11. Determine what time the next test will be performed and inform the patient.
12. Remind the patient that they cannot consume anything but water during the testing period.
13. Repeat the collections at exactly the selected collection time and transport as quickly as possible to the laboratory until every sample has been collected.
14. Observe the patient throughout the procedure and answer any questions they may have.
15. When all the testing has been completed, thank the patient and inform them that they are finished. Inform them of when they should receive their results.

## *Glucose Tolerance Test*

During a patient's pregnancy, there may be concerns that they have a type of diabetes that exists only during pregnancy. After the birth of the child, the patient's body will return to its normal glucose levels. This diabetic condition during pregnancy is called gestational diabetes. To determine if the patient has this condition, their ordering provider may order a glucose tolerance test. This test requires the patient to adhere to a specific diet prior to the test and avoid consuming anything but water while the test is being performed.

Follow the routine steps for performing a GTT. There are a few changes as this test is for gestational diabetes. The procedure is sometimes only performed for over an hour, which differs from a routine glucose tolerance test. If this test comes back with elevated glucose levels, the test may continue on to the 3-hour GTT. The glucose beverage is typically 50 g of the glucose solution. Every step should be followed as instructed to help ensure accurate test results.

### Two Hour Post-Prandial Glucose Tolerance Test

Another glucose tolerance test that may be ordered is called a 2-hour post prandial test. This test is performed after a patient has eaten to help determine how their body processes glucose. This blood test is performed 2 hours after a patient has eaten. Before the patient eats, remind them to record what time that they finish their meal. Perform the test as close to the 2-hour period as possible to help ensure accurate test results. Record the procedure, confirming that the times recorded accurately reflect both the time the patient ate and when the collection was performed.

**Tolerance Testing Timing**

Timing is very important for the accuracy of glucose tolerance tests. Always document the exact time the procedure started and perform all blood and urine collections at the scheduled time.

## *Lactose Tolerance Testing*

A patient undergoing side effects after they consume dairy products may be experiencing lactose intolerance due to the fact that their bodies do not make lactase, the enzyme that helps break down lactose. Lactose is a sugar unique to dairy products. To determine if a patient is intolerant to lactose, a lactose tolerance test may be ordered. This test is performed similarly to a glucose tolerance test, but instead of glucose, you will be administering lactose. Some providers may require that a GTT is performed 24 hr before the lactose tolerance test. Confirm that the results from the GTT are in range before performing the lactose tolerance test.